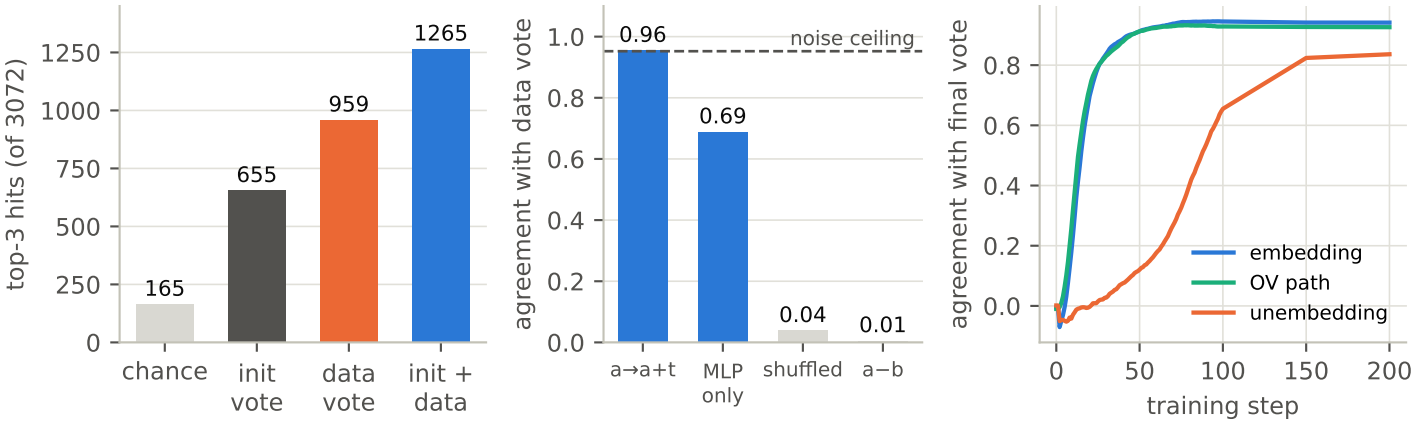
(a) Held-out 32 × 32 factorial
top-3 hits (of 3072)
1250
1000
750
500
250
0
165
655
959
1265
chance
init vote
data vote
init + data
(b) What casts the data vote
agreement with data vote
1.0
0.8
0.6
0.4
0.2
0.0
0.96
noise ceiling
0.69
0.04
0.01
a→a+t
MLP only
shuffled
a−b
(c) It forms within ~20 steps
agreement with final vote
0.8
0.6
0.4
0.2
0.0
embedding
OV path
unembedding
0
50
100
150
200
training step